Project Profile: Agar Aisa Ho Jai — Radeon Story Studio

AMD AI DevMaster Hackathon — Track 1: Multimodal AI
Participant: Ankur Sawhney
Public source snapshot: eba2f1544f03fbc6e7ad6dd137138147e5be3f26

1. Project background

Children’s educational media can fail even when AI-generated visuals look polished. A scene may show an ambiguous action, place a child too close to a hazard, or drift in character identity across panels. This project treats media generation and acceptance as separate stages, and records evidence for both.

Agar Aisa Ho Jai — Radeon Story Studio is a local Radeon/ROCm content workflow that turns a short educational scenario into a reviewable storyboard and child-safe visual lesson. The demonstration scenario is a three-step safe-charger lesson: notice the damaged cable, stop and step back, and tell an adult.

2. Target users and scenarios

• Educators and child-safety content creators producing short instructional stories.

• Small production teams that need traceable model/runtime evidence instead of one-off black-box generation.

• Reviewers who must separate “image generation succeeded” from “content is safe and semantically correct.”

• Creators who prefer local/private GPU inference rather than depending on a closed online generation API for the core workflow.

3. System architecture

Stage	Purpose	Radeon/ROCm role
Transcript / sealed manifest	Defines allowed content, characters, and actions.	Input contract for local workloads.
Qwen + vLLM storyboard planning	Generates constrained JSON storyboard and second-pass textual safety review.	Local inference served on AMD Radeon GPU through ROCm-backed vLLM.
SDXL Turbo image generation	Creates modular candidate story panels and accepted lesson-card assets.	PyTorch/Diffusers image inference on Radeon ROCm device.
Validation and review	Checks technical pass, safety, semantics, continuity, hashes, and review status.	Uses recorded GPU evidence; fails closed before publication.

4. Models and algorithms

SDXL Turbo image generation: immutable model revision, 512×512 candidates, warm-up separated from measured inference, text-to-image and image-to-image passes, and fail-closed technical/safety/semantic/continuity gates.

Qwen storyboard planning: locally served Qwen3-Coder 30B-A3B GGUF through vLLM, constrained JSON schema, enumerated actions/backgrounds/emotions, second-pass textual safety review, deterministic Pillow preview rendering, FFmpeg animatic, checksums, and motion evidence.

5. AMD Radeon / ROCm adaptation

• PyTorch reports a non-empty HIP version and uses its cuda:0 API over ROCm.

• Runtime preflight verifies an AMD Radeon device and a real tensor operation.

• Component dtypes are observed rather than guessed from configuration.

• Model warm-up is excluded from measured warm-throughput metrics.

• Evidence records gfx1100, ROCm-enabled PyTorch, GPU memory, timings, token throughput, and artifact hashes.

• Execution packages and returned results are content-addressed and verified.

6. Measured results

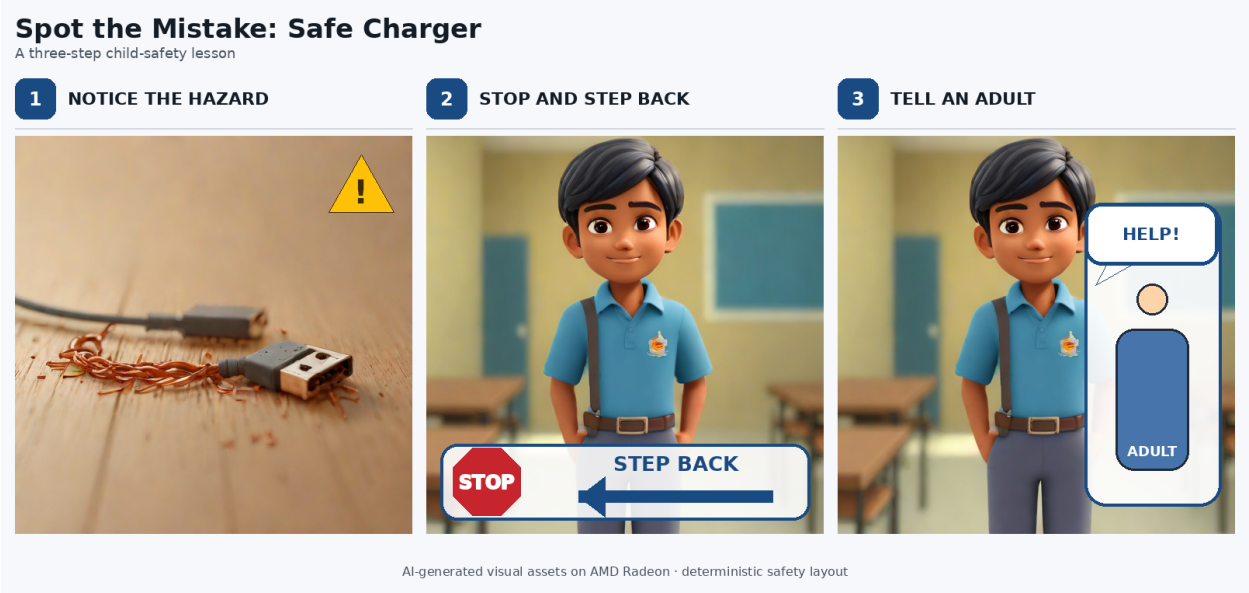
Workflow	Measured evidence
SDXL Story Studio	8 candidates; 10.504696 seconds measured candidate phase; 0.368234-second median candidate latency; 45.693848 candidates/minute; ~10.45 GB peak allocated GPU memory.
Qwen Storyboard Factory	7-scene plan, 6 cast members; 84.264 completion tokens/second for planning; 58.389 completion tokens/second for textual safety review; 7/7 motion scenes and semantic action checks validated.

7. Innovation and practical value

The main contribution is the evidence-backed boundary between technical generation and editorial acceptance. Failed or ambiguous scenes remain rejected even when GPU inference succeeds. Unsafe raw candidates are represented by hashes and rejection reasons, not published as artwork.

This makes the workflow useful for small creators: it combines local GPU inference, reproducible evidence, and child-safety review discipline without relying on closed online APIs for the core multimodal work.

8. Accepted demonstration output



Accepted safe-charger lesson card assembled from fictional Radeon-generated visual assets and deterministic safety layout.

9. Limitations and next steps

• The Qwen safety review covers structured text; it is not claimed as a complete multimodal judgment of every rendered frame.

• The deterministic animatic is a preview workflow rather than polished production video.

• Final publication still requires manual review of every visual, model, font, and asset-rights condition.

• Future work: local multimodal frame review, stronger pose/continuity control, and richer child-friendly story formats.

10. Publication boundary

This public snapshot was built from an explicit allowlist. It excludes private real-person/children corpora, Codex context archives, nested repositories, local sessions, credentials, personal paths, model weights, rejected raw media, historical campaigns, unrelated provider adapters, and uncommitted work.

11. Supporting materials

• README with environment, setup, validation, package preparation, and measured-result summary.

• Project Profile PDF, PowerPoint presentation, demo script, PR title and PR body draft.

• Accepted lesson-card image plus machine-readable evidence for SDXL and Qwen runs.

• Focused tests for Radeon runtime discovery, cloud execution, session transport, SSH execution, and story-studio validation.

12. Reproducibility notes

The repository intentionally does not bundle large model weights. Judges can inspect the source and evidence without downloading multi-gigabyte assets, while the README documents the expected ROCm-enabled Python environment and execution commands. The included manifest and SHA-256 file make the submitted snapshot auditable at file level.